East Swindon Development Group

Eastern Villages Swindon (South Marston)

DORMOUSE SURVEY REPORT

November 2022

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1.0 NON-TECHNICAL SUMMARY

1.1 Suitable dormouse habitat was present within the site in the form of mixed plantation woodland, hedgerows and mature trees.

1.2 Monthly presence/likely absence dormouse surveys were undertaken between June and November 2022 inclusive; evidence of dormice was not recorded on any survey occasions.

1.3 Dormice are not considered to represent a constraint to the development

2.0 INTRODUCTION

2.1 This report has been produced by FPCR Environment and Design Ltd. on behalf of a consortium comprising Hallam Land Management Ltd., Taylor Wimpey and Hannick Homes. It provides the results of hazel dormouse *Muscardinus avellanarius* surveys conducted at land in South Marston, near Swindon.

Site Context

2.2 In broad terms, the site forms a parcel of land lying between the small village of South Marston to the north, and the A420 and the Great Western railway line to the south. The site is primarily under agricultural management at present, and includes arable crops, turf production and pastoral fields bisected by hedgerows.

3.0 HAZEL DORMOUSE LEGISLATION AND ECOLOGY

Legislation

3.1 The hazel dormouse is listed under Annex IVa of the EC Habitats Directive and as a result is covered by Section 41 of the Conservation of Habitats and Species Regulations 2017. It is also protected under the Wildlife and Countryside Act 1981 (as amended). Taken together, these make it an offence to:

- deliberately capture or intentionally take a dormouse;
- deliberately or intentionally kill or injure a dormouse;
- to be in possession or control of any live or dead dormouse or any part of, or anything derived from a dormouse;
- damage or destroy a breeding site or resting place of a dormouse;
- Intentionally or recklessly obstruct access to any place that a dormouse uses for shelter or protection;
- intentionally or recklessly disturb a dormouse while it is occupying a structure or place that it uses for shelter or protection; and
- deliberately disturb any dormouse in particular any disturbance which is likely to:
    - impair their ability to survive, breed, reproduce or to rear or nurture their young; or in the case of hibernating or migratory species, to hibernate or migrate; or
    - to affect significantly the local distribution or abundance of the species to which they belong.

3.2 Although the law provides strict protection to hazel dormice, it also allows derogation from this legal protection under Section 53 of the Conservation of Habitats and Species Regulations 2017 through the issuing of European Protected Species (EPS) licences for development works. These licences in England are currently determined by Natural England (NE).

3.3 Where a lawful operation is required to be carried out, which is likely to result in one of the above offences, an EPS licence may be obtained from NE to allow the operation to proceed. However, The Dormouse Conservation Handbook[1] states that the decision for a licence is to be made by the consultant and client; it also states that *"No licence is required if the proposed activity is unlikely to result in an offence"*. It also stipulates that, "…*if the proposed activity can be timed, organised and carried out to avoid committing offences, then no licence is required*."

3.4 The hazel dormouse is listed as a Species of Principal Importance in the Natural Environmental & Rural Communities Act (NERC Act, 2006) and as a result public bodies must have regard to its conservation when carrying out their duties.

Ecology

3.5 The hazel dormouse is a largely nocturnal rodent, found in deciduous woodland and hedgerows. It is mostly arboreal, spending much of its time climbing amongst tree branches in search of food and rarely coming to ground. Dormice need a variety of trees and shrubs within their habitat to provide the variety of food sources throughout the year. Food sources include flowers, pollen, fruits, insects, seeds and nuts.

3.6 Dormice are able to lower their body temperature and become torpid, they use this strategy during the spring and summer, to wait out periods where food supply is short or bad weather prevents foraging. Dormice hibernate during the winter months within nests on the ground under leaf litter or within hedgerow bases. In the spring, dormice become active again around April and May.

3.7 During the day dormice sleep in a nest. This can be in a hollow tree branch, a deserted bird nest or nest box, and in specific dormouse nest boxes or nest tubes.

3.8 Dormice live at low population densities and have home ranges that stretch between 1.5 to 1.7ha dependant on sex, habitat quality and habitat shape (longer home ranges are often observed in individuals confined to hedgerows). If habitats are suitable and fall within known areas of dormice occupation, then regardless of habitat size, surveys for dormice should be undertaken, unless there are limited historical linkages. Dormice can live up to five years in the wild, longer than other small mammals of a comparable size. During the breeding season up to five young are born usually between June to September.

[1] Bright, Morris & Mitchell-Jones 2006. *The Dormouse Conservation Handbook*. English Nature, Peterborough.

4.0 METHODOLOGY

4.1 As part of the Desk Study for the Ecological Appraisal[2] information was sought from Wiltshire & Swindon Biological Records Centre for the presence of dormouse records within 1km of the survey site.

4.2 Further inspection, using colour 1:25,000 OS base maps (www.ordnancesurvey.co.uk) and aerial photographs from Google Earth (www.maps.google.co.uk) was also undertaken in order to provide additional context and identify any features of potential importance for nature conservation in the wider countryside.

Field Surveys

General Habitat Suitability

4.3 The hedgerows, woodland and trees present within the survey area were assessed regarding their suitability to support hazel dormouse, with particular consideration given to species composition, structure and connectivity to the wider area.

Nesting Tubes

4.1 Presence/absence surveys have been undertaken in accordance with current good practice guidelines (Bright *et al.* 2006)[3]. Surveys involve placing standard dormouse nest tubes every 20m in suitable habitat, approximately 1.5m above ground. Survey effort is scored using the standard method described previously by Chanin and Woods (2003)[4].

4.2 An index of probability is used to calculate the likelihood of finding dormouse based on seasonal variation in dormouse activity and the number of tubes deployed in suitable habitat, where a standard of 50 tubes equals 1. 50 tubes therefore placed across the whole season would score 26 (sum of April – November) (*Table 1*). A score of 20 or more is considered a suitable survey effort.

Table 1: Dormouse survey index table

Month	Index of Probability
April	1
May	4
June	2
July	2
August	5
September	7
October	2
November	2

[2] FPCR 2020. *Land South of Leamington Road, Southam Ecological Appraisal*
[3] Bright, P., Morris, P. and Mitchell-Jones, T. 2006. *The Dormouse Conservation Handbook*. English Nature, Peterborough.
[4] Chanin, P., and Woods, M. 2003. *English Nature Report No. 524 on nest tube surveying*. English Nature, Peterborough.

4.3 A total of 120 dormouse nest tubes were deployed in linear features suitable for dormice on the 3rd May 2022 (Figure 1). The tubes were checked on six occasions (Table 2).

Table 2: Dormouse survey occasions

Date	Surveyors
May 3rd – Nesting tube deployment	Zoe Barnes and Edward Peters
June 6th	Tiana Thomas and George Allen
July 4th	Suzy Mitchell and George Allen
August 2nd – August 3rd	Katie Oliver and Laura Pirateque
September 29th	Caitlin Page and Zoe Barnes
October 20th	Caitlin Page and Joe Freemantle
November 25th	Joe Freemantle

Limitations

4.4 Nesting tubes were deployed at a density greater than the recommended minimum density and were checked the appropriate number of times. No tubes were lost or damaged during the survey period.

5.0 RESULTS

Desk study

5.1 One record of hazel dormouse was returned from the Wiltshire & Swindon Biological Records Centre consultation from within a 2km radius of the site (dated 2013).

Field Survey

Habitat

5.2 On-site field boundaries are dominated by hedgerows, which frequently include mature standard trees. Typically, hedgerows were generally dominated by hawthorn with abundant blackthorn *Prunus spinosa*, elm *Ulmus* sp. and dog rose *Rosa canina*. Many of the hedgerows are tall and thick containing a number of mature trees, although some are variable in quality, largely due to current management practices.

5.3 Opportunities for foraging throughout the year would be provided by the variety of fruit and nut bearing plants. Hawthorn flowers provide an important food source in the spring, and hawthorn fruits are eaten occasionally. Blackthorn kernels are also eaten, however denser blackthorn thickets are generally avoided.

5.4 The hedgerows provide good connectivity within the site but habitat connectivity to wider habitats is limited due to the presence of the industrial estate, railway line, and Swindon town to the south, north and west respectively.

5.5 Results of the dormouse survey occasions are detailed in Table 3.

Table 3 Dormice Tube Checking Dates and Results

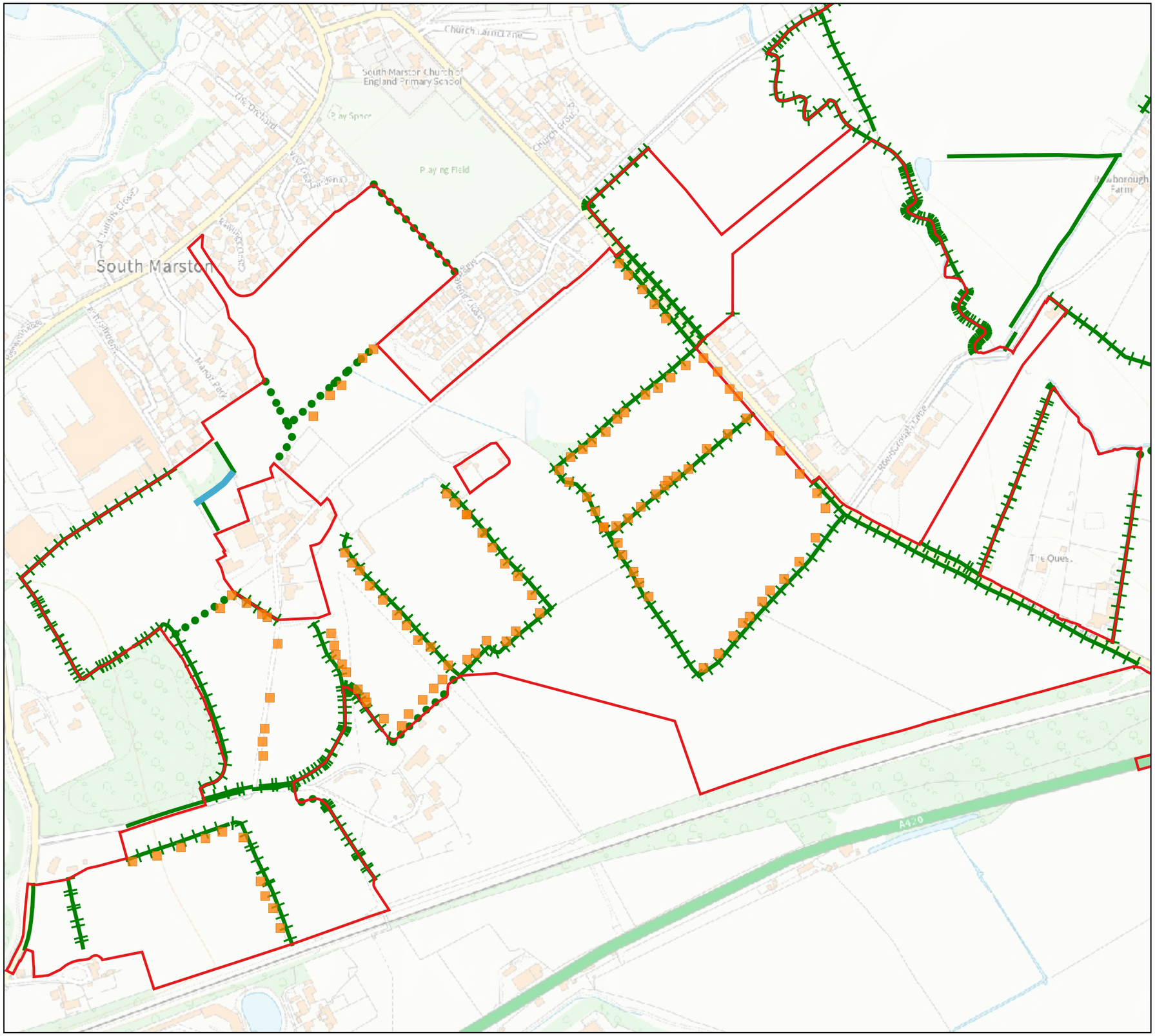
Survey	Date	Dormouse evidence recorded
1	June 6th	None
2	July 4th	None
3	August 2nd – August 3rd	None
4	September 29th	None
5	October 20th	None
6	November 18th	None

6.0 DISCUSSION

6.1 One record of hazel dormice returned by the desk study consultation from within the 2km search radius (dated October 2013.

6.2 Due to the presence of suitable habitats throughout the site, presence/likely absence surveys for hazel dormice were carried out between June and November 2022 in accordance with the good practice guidelines. The survey effort achieves the recommended score of 20.

6.3 Despite suitable habitat being present, no evidence of dormice was found during the surveys and therefore the likely absence of dormice on site can be established. As such hazel dormouse is not considered a constraint to the development.



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Key

- Dormouse Nesting Tube Location
- Site Boundary

Hedgerows

- Native Hedgerow with trees
- Native Hedgerow
- Line of Trees
- Hedge Ornamental Non Native

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South Marston,
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DORMOUSE NEST TUBE LOCATIONS

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